

Personalized recommendation of entertainment robots in fine arts education based on human–computer interaction and data mining

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ABSTRACT

In the current digital entertainment environment, the art learning mode of human–computer interaction can effectively enhance students' interest. Entertainment robots, as a novel personalized interaction tool, can provide a more intelligent and entertaining way for art teaching. This article analyzes the personalized recommendation of entertainment robots in art education based on human–computer interaction and data mining. On the basis of collecting a large number of fine art education resources and organizing and labeling them, data mining technology is used to extract user background information and interest preferences. Information retrieval technology is used to filter out content related to user interests from the massive art education resources based on the extracted user interests. We also use recommendation algorithms to sort and personalize the recommended content based on the user's historical behavior and other user evaluations, in order to improve the accuracy of recommendation results and user satisfaction. The effectiveness and practicality of this personalized recommendation model were verified through user surveys and data analysis. The experimental results indicate that this model can quickly and accurately recommend suitable fine art education content for users, improving their learning effectiveness and experience.

1. Introduction

Folk fine art is an important component of traditional Chinese art, originating from the creativity and practical life of the working people [1]. It expresses people's pursuit and love for a better life through artistic forms. There are various forms of Folk fine art, including painting, sculpture, ceramics, embroidery, woodworking, etc. Each form carries rich cultural connotations and ethnic characteristics. In the development process of Chinese history and culture, Folk fine art resources have accumulated rich, including many unique artworks and techniques [2]. Folk fine art is not only a form of artistic creation, but also an important way for working people to learn and cultivate art. By participating in the creation and appreciation of Folk fine art, people can cultivate their ability to perceive beauty, improve their aesthetic level, and enrich their spiritual life. Folk fine art also plays an important role in cultural inheritance and social connection [3]. Through the inheritance and development of traditional techniques, Folk fine art conveys rich historical and cultural information, connecting the past and present, and shaping the artistic image and cultural characteristics of the nation [4]. In modern society, with the development of the economy and the display

of cultural diversity, Folk fine art and culture with traditional cultural characteristics gradually fade away. In the current cultural context, we should actively explore the cultural resources of Folk fine art and attach importance to the protection and inheritance of Folk fine art [5]. Firstly, exhibitions, art festivals, and other forms can be held to showcase Folk fine art works to the public, allowing people to better understand and appreciate the charm of Folk fine art. We can also strengthen training and support for Folk fine art practitioners, encouraging them to create more innovative and artistic works. Folk fine art can also be included in the art education curriculum of schools, allowing students to have more exposure and understanding of Folk fine art, and cultivating their sense of pride and confidence in their own national culture [6]. By actively exploring and inheriting Folk fine art and culture, not only can we protect and inherit folk culture with Chinese characteristics, but we can also provide more art education content and forms for the education field.

Art education plays an important role in cultivating students' creativity, aesthetic ability and artistic expression ability. However, there are some problems in the traditional art education, such as the lack of personalized teaching and the challenge of students' interest guidance.

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These problems may lead to a decline in students' interest in art education and poor learning results. In recent years, the rapid development of human-computer interaction and data mining technology has brought new opportunities for art education. As an innovative teaching tool, entertainment robots bring new possibilities to art education through the interaction with students and the introduction of entertainment elements. Entertainment robots can simulate the role of a human teacher, interact with students and provide a personalized learning experience. It can tailor art education content and activities according to students' interests, learning progress and learning style. For college students, a better understanding of the culture and connotation of Chinese Folk fine art is beneficial for enhancing their sense of identification and confidence in their own national culture [7]. The inheritance and development of Folk fine art can also bring more cultural innovation and economic benefits to society. The basic principle of the personalized recommendation method for Folk fine art education designed in this article is to use data mining and information retrieval technology. Through the personalized recommendation method, users can obtain personalized recommendation content related to Folk fine art teaching according to their interests and needs [8]. Based on human-computer interaction and data mining technology, the personalized recommendation system of entertainment robot can analyze students' personal background information and learning data, find students' interests and needs, and provide corresponding art education resources and guidance according to these information. Through personalized recommendations, entertainment robots can help students better master art knowledge and skills, and stimulate their creativity and artistic interest. Through the combination of entertainment robot and personalized recommendation technology, students can be provided with a richer and more diverse art education experience and improve their learning enthusiasm and learning effect, which has important practical significance for promoting the development of art education and improving students' artistic literacy.

2. Related work

The literature adopts software engineering ideas to optimize the teaching management system of art courses [9]. Through the exploration, practice, and research of various technologies in software engineering, the literature has found various flow charts that meet the needs of art course teaching management [10]. These flowcharts include but are not limited to performance requirements, professional work requirements, data requirements, and usage requirements. The literature has designed a simulation system for art teaching assistance, which includes multiple business modules and covers the main teaching activities [11]. By establishing an art library, students can easily access various art resources, such as painting works, art history materials, etc., thereby expanding their artistic horizons. Another important module is online teaching. Through the online environment, teachers can teach remotely and students can participate in learning anytime and anywhere. This makes teaching more flexible and convenient, while also increasing student engagement.

The literature has studied the shortcomings of current recommendation algorithms and models [12]. To address this issue, the hybrid personalized recommendation technology introduces demographic information analysis technology to cluster users. By analyzing demographic information such as age, gender, and region of users, similar users are grouped together to form a user clustering group [13]. On the basis of clustering user groups, literature uses collaborative filtering algorithms for resource recommendation. The collaborative filtering algorithm analyzes the historical behavior and interests of users, identifies other users with similar interests, and then recommends the resources that these users like to the target user [14]. By introducing demographic information and user clustering groups, literature can more accurately identify other users with similar interests and recommend their favorite resources to target users. This personalized

recommendation method can better meet the needs and interests of users, and improve the quality of recommendations. The literature provides in-depth analysis of experimental data [15]. Through the statistics and comparison of user evaluation information, it was found that the personalized resource recommendation model based on hybrid recommendation technology can effectively solve the problem of sparse user evaluation information and fill the gap in evaluation information. This further proves the advantages of this model in improving recommendation accuracy.

3. Research on personalized information retrieval algorithms based on web data mining

3.1. Dynamic clustering technology

In the process of personalized recommendation, entertainment robots use data mining technology to analyze students' learning behaviors and learning preferences, so as to provide them with customized art education resources and activities. Entertainment robots can interact with students to learn about their interests and art skills. By collecting students' personal background information and learning data, the entertainment robot can build a student's learning profile, including data on their interests in the field of fine arts, learning progress and learning style. Using data mining technology, entertainment robots can analyze students' learning behavior and learning outcomes. By evaluating the student's work and learning record, the entertainment robot can understand the student's strengths and areas for improvement. In this way, the entertainment robot can recommend suitable art education resources for students according to their individual needs, such as painting tutorials, art appreciation activities, art competitions, etc. Entertainment robots can also interact with students by means of human-computer interaction. Through question-and-answer conversations with students, entertainment robots can learn how students understand and feel about art. Based on these interactive data, the entertainment robot can adjust the personalized recommended content according to the feedback and preferences of students, providing educational resources that are more tailored to the needs of students.

The characteristics of web topology include link relationships and the structure of network diagrams. Link relationships refer to the hyperlinks between web pages, while the structure of a network diagram refers to the connection and hierarchical structure of web pages. These characteristics reflect the semantic relationships and correlations between texts, but are often overlooked in traditional vector space models. In order to better utilize the characteristics of web topology, it can be considered to cluster this information together with text content.

For each page W_i , use formula (1) to calculate the total number of times that page has been accessed by all users per unit time:

$$T_i = \sum_{k=1}^m N(W_i, U_k) \quad (1)$$

Then,

$$E(W_i, U_j) = \frac{N(W_i, U_j)}{T_i} \quad (2)$$

To measure the correlation between page W_i and user U_j , use it as feature data and obtain a raw data matrix $B_{n \times m}$ with n rows and m columns, as shown in formula (3):

$$B_{n \times m} = \begin{bmatrix} E_{11} & \cdots & E_{1m} \\ \vdots & & \vdots \\ E_{n1} & \cdots & E_{nm} \end{bmatrix} \quad (3)$$

Convert the original data matrix B_m into a fuzzy similarity matrix R . Use the maximum minimum method to process the original data matrix $B_{n \times m}$ and obtain the fuzzy similarity matrix R . By using this matrix, determine the degree of similarity between each page, that is, their

similarity in relevance.

$$r_{ij} = \frac{\sum_{k=1}^m \min(B_{ik}, B_{jk})}{\sum_{k=1}^m \max(B_{ik}, B_{jk})} \quad (4)$$

An effective feature item set should be able to reflect both the target content and distinguish the target from other documents, and construct a feature item weight evaluation function, as shown in formulas (5) and (6):

$$\text{Weight}(w) = tf_{ik} \times IDF_i \quad (5)$$

$$IDF_i = \log\left(\frac{N}{n_k} + 1\right) \quad (6)$$

If the length of an entry is considered and standardized, formula (7) is obtained:

$$\text{weight}(w) = \frac{tf_{ik} \times \log\left(\frac{N}{n_k} + 1\right)}{\sqrt{\sum_{k=1}^n (tf_{ik})^2 \times \log^2\left(\frac{N}{n_k} + 1\right)}} \quad (7)$$

Through this processing, the weight of entries can be more accurately evaluated, thereby better reflecting their importance in the feature set.

3.2. The main process model of information retrieval

The model of the information retrieval process is shown in Fig. 1:

Through natural language processing technology, entertainment robots can recognize and represent the semantic content of documents, transforming it into a form that computers can understand and process. It can describe and analyze artistic works, extract and display information such as artist's inspiration, creation intention, style and theme of works. At the same time, the entertainment robot can also identify the author, editor and other contextual attributes of the work, and share and discuss this information with students. Through the application of text models, entertainment robots can help students better understand and analyze artistic works, and deeply explore the cultural connotation and artistic value behind them. Students can discuss the meaning, expression and artistic elements of the work through dialogue and communication with the entertainment robot, so as to cultivate their art appreciation

ability and critical thinking. By analyzing students' learning history and interest preferences, the entertainment robot can select the most suitable text materials from a large number of teaching resources, including art criticism, relevant research and lectures by famous artists. These personalized recommendations can help students delve deeper into their areas of interest in the arts and further enhance their learning effectiveness and motivation.

Among them, text model refers to the indexing of documents, that is, the recognition and representation of document content. This includes extracting and representing the semantic content and contextual attributes of the document (such as author, editor, etc.). The purpose of a text model is to transform the content of a document into a form that can be understood and processed by a computer for subsequent retrieval and sorting.

The query model refers to the acquisition and representation of user demand information. The query model is used to transform user needs into a form that computers can understand and process, in order to match and retrieve with text models. The query model can include the keywords entered by the user, their logical relationships, and other relevant information.

The sorting function refers to the function of sorting the detected literature according to its relevance. The sorting function determines the relevance of literature based on its matching degree with the query.

Index structure refers to the index structure formed by indexing literature databases. The index structure is typically represented by inverted documents, which support fast search in large amounts of data by indexing the keywords of the documents. The inverted document index can quickly locate documents containing that keyword based on the keyword.

Feedback correction is the process of expanding and optimizing query expressions based on search results. By analyzing user feedback on search results, query expressions can be modified to better meet user needs and improve system performance.

In the Boolean model, to measure the similarity between literature d_j and query q , formula (8) can be defined:

$$\text{sim}(d_j, q) = \begin{cases} 1 & , \exists q_{cc} I(q_{cc} \in q_{df}) \wedge (\forall k_i, g_i(d_j) = g_i(q_{cc})) \\ 0 & \text{other} \end{cases} \quad (8)$$

If the result is equal to 1, it indicates that literature d_j is related to query q . The Boolean model uses an inverted index storage structure,

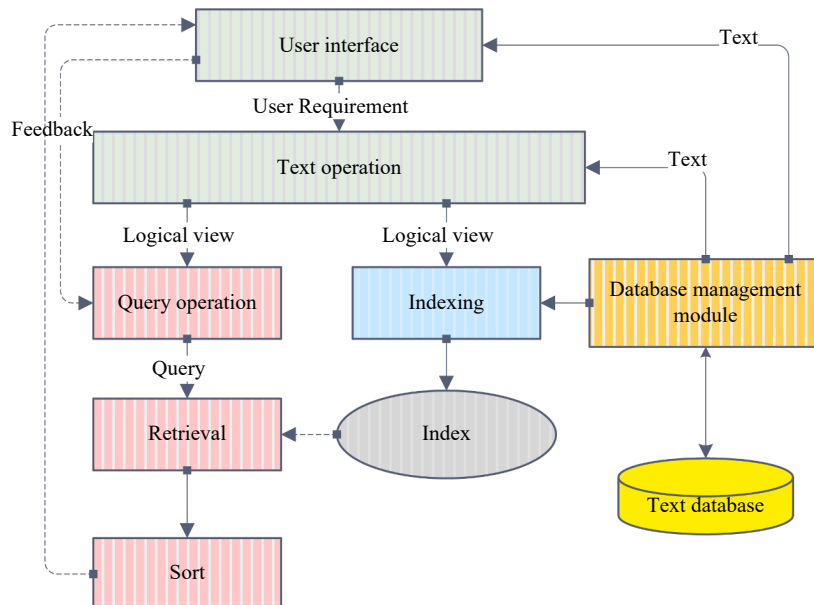


Fig. 1. Model diagram of information retrieval process.

which enables convenient and efficient collection operations. But the Boolean model also has some shortcomings. Firstly, it is difficult to control the accuracy of the search results because it can only return literature that meets the query criteria and cannot perform correlation sorting on the results. Secondly, Boolean models do not support local matching between queries and literature, which means they cannot consider information such as the position and frequency of keywords.

To address the shortcomings of Boolean models, similarity calculation between document vectors and query vectors can be introduced. The similarity can be expressed using formula (9):

$$\text{sim}(d_j, q) = \frac{d_j \cdot q}{|d_j| \times |q|} = \frac{\sum_{i=1}^t w_{ij} \times w_{i,q}}{\sqrt{\sum_{i=1}^t w_{ij}^2} \times \sqrt{\sum_{i=1}^t w_{i,q}^2}} \quad (9)$$

According to Bayesian theorem, the similarity formula can be further expanded to obtain formula (10):

$$\text{sim}(d_j, q) = \frac{P(d_j|R) \times P(R)}{P(d_j|\bar{R}) \times P(\bar{R})} \quad (10)$$

By calculating similarity, literature can be sorted for relevance and the most relevant literature can be returned as the search result based on the magnitude of similarity. This can improve the quality and accuracy of search results. Similarity calculation can also consider information such as the position and frequency of keywords, in order to achieve more accurate matching and sorting.

3.3. Weighted association rule mining technology

The classic Apriori algorithm discovers frequent itemsets by analyzing the frequency of items in the database. But this algorithm hides a prerequisite that each item is equally important in data mining. Actually, different items may have different values and importance. In real life, we often encounter some low frequency but very important items. Mining association rules solely based on the frequency of items in the database may overlook these important items.

Therefore, it is very important to consider the value of items when studying association rules. A weighted association rule algorithm can be used, where each item in a transaction is assigned a weight value to represent the user's interest in the item. The weighted association rule algorithm in this article uses two factors to determine the weights of items: the access time of learning resources and the access frequency of learning resources. The access time of learning resources refers to the duration of user access to a certain item. If a learning resource is accessed by a user for a long time, it can be considered more valuable to the user. The frequency of access to learning resources refers to the number of times a user accesses a certain item. If a learning resource is frequently accessed by users, it can be considered more attractive and important to the user. Therefore, using access time and frequency as components of the weights can more accurately reflect users' interest and importance in learning resources. When mining association rules, these weights can be used to adjust the thresholds of support and confidence, in order to discover more valuable association rules.

The algorithm is mainly divided into two steps: in the weighted association rule mining stage of resource clustering, the algorithm clusters each resource and mines weighted association rules in each cluster. This means that the algorithm will be weighted based on the user's interest in learning resources to obtain more accurate association rules. By considering the weights of items, the algorithm can discover more meaningful and useful association rules. These association rules can reveal the correlation between learning resources and users' learning preferences, providing important information for recommendation systems. During the resource recommendation phase of the recommendation system, active user sessions are passed into the system. The recommendation system will search for the most similar association rules based on the user's active session. The calculation of similarity can

be based on factors such as the degree of overlap between association rules and learning resources in active conversations, as well as sequence patterns. The recommendation system will consider the weighted confidence of association rules to determine the recommended learning resources. Weighted confidence can reflect the reliability and importance of association rules, thereby helping recommendation systems choose learning resources that are more likely to meet user interests.

The weight pattern uses the access time and frequency of learning resources to evaluate their importance in transactions. The access time for learning resources is chosen because users typically spend a considerable amount of time on useful and interesting learning resources. However, in order to consider the impact of the amount of learning resource content on actual access time, it is more reasonable to use unit time. Formula (11) provides a method for calculating this weight value:

$$\text{Duration}(p) = \frac{\frac{\text{Total Duration}(p)}{\text{size}(p)}}{\max_{q \in T} \left(\frac{\text{Total Duration}(p)}{\text{size}(p)} \right)} \quad (11)$$

In formula (11), it is assumed that the speed at which each user accesses learning resources is similar, and the speed at which people browse web pages and obtain learning resources varies little. Therefore, the time users spend accessing learning resources is directly proportional to the useful information they obtain from them.

The higher frequency of access to learning resources means that users have a higher level of interest in the learning resources. But in order to more accurately evaluate the importance of learning resources, it is also necessary to consider the degree of entry of the page where the learning resource is located, that is, the number of links imported into the page. Because pages with high entry rates are more likely to be frequently accessed, it is necessary to include entry rates in the weight calculation.

Formula (12) provides a method for calculating the frequency of access to learning resources.

$$\text{Frequency}(p) = \frac{\text{Number of visit}(p)}{\sum_{q \in T} \text{Number of visit}(p)} * \frac{1}{\text{Indegree}(p)} \quad (12)$$

The weight of learning resources is defined as formula (13), which evaluates the importance of learning resources in user transactions by taking into account their access time and frequency through product analysis.

$$\text{Weight}(p) = \text{Frequency}(p) * \text{Duration}(p) \quad (13)$$

Recommend the most suitable learning resources for users according to their interests and needs. The weight of the item set X, denoted by w(X,t):

$$w(X, t) = \begin{cases} \min(w(p_1, p_2, \dots, p_k)) & X \subseteq t \\ 0 & X \not\subseteq t \end{cases} \quad (14)$$

The weights of things are used to calculate the support of itemsets. Calculate the average weight of all items included in a transaction as the weight of the transaction. The calculation formula is shown in formula (15):

$$w(t_k) = \frac{\sum_{i=1}^{|t_k|} w(p_i)}{|t_k|} \quad (15)$$

The weighted support wsp(x) of itemset X is defined as formula (16) to measure the frequency of itemset X appearing in transactions. It is the product of the weight of the itemset X and the weight of the transaction:

$$\text{wsp}(x) = \frac{\sum_{t_i \in T} w(t_i) * w(X, t_i)}{\bar{w} * \sum_{k=1}^{|t_k|} w(t_k)} \quad (16)$$

The weighted confidence of weighted association rules is used to measure the reliability of the rules. The definition formula is shown in (17):

$$wconf(X \Rightarrow Y) = \frac{wsp(X \cup Y)}{wsp(X)} \quad (17)$$

Evaluate the reliability of rules by calculating weighted confidence and determine whether they can be used as part of association rules.

3.4. Data information classification results

According to the experimental results, the use of the classifier for classification is shown in Table 1.

According to the data in Table 1, the number of training documents for the social fire painting category is relatively small, and the range of feature words in this category is relatively narrow. Due to insufficient training samples, the classifier may not learn the features of the social fire painting category well, resulting in poor classification performance. In order to improve the classification performance of social fire painting categories, it is possible to consider increasing the number of training samples for that category, or further analyzing and extracting specific features of social fire painting categories to improve the performance of the classifier.

According to the test results shown in Fig. 2, it can be seen that the number of training samples has a direct impact on both macro and micro F values. As the number of training samples increases, the performance of the classifier gradually improves.

According to Fig. 2, it is observed that when the number of training sample sets exceeds 5500, the Macro-F₁ value of the classifier reaches a stable state, while the Micro-F₁ value is still increasing, but the growth rate becomes slow. This indicates that as the number of training samples increases, the overall performance of the classifier is stable, but there is still room for improvement in the classification performance of small categories. Through experiments, it was also found that in the classifier designed in this article, the more feature items, the better. The experimental results show that when about 50 feature items are selected for each document, the local optimal classification effect can be achieved. This indicates that excessive feature selection may introduce noise or redundant information, thereby reducing the performance of the classifier. Therefore, in practical applications, it is necessary to make a balance between the number of training samples and the selection of feature items in order to obtain the best classifier performance.

4. Research on personalized recommendation methods for Folk fine art education

4.1. Personalized recommendation model structure

The personalized recommendation method for Folk fine art education is one of the key components of web-based personalized information services. This method provides personalized Folk fine art teaching services to learners by analyzing their usage behavior and characteristics. Personalized retrieval provides learners with accurate search results for Folk fine art education resources by analyzing their search behavior and interest preferences. Personalized recommendation recommends personalized Folk fine art education content to learners based on their historical learning behavior and interest preferences. Personalized user

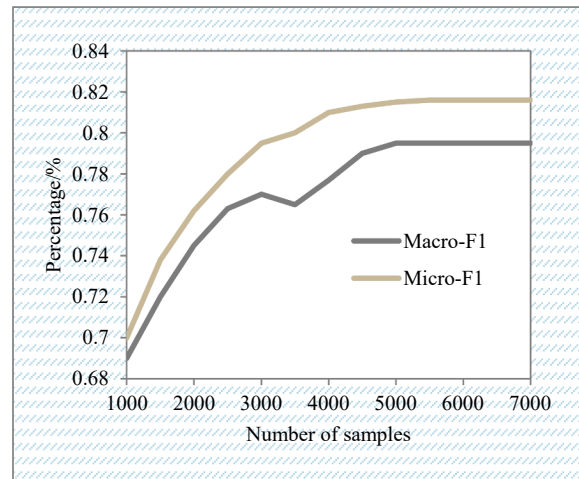


Fig. 2. Curve Chart of Macro-F1 and Micro-F1 Changing with the Number of Training Samples.

interface adjustment adjusts the display and interaction methods of the system interface based on learners' usage habits and feedback, in order to improve learners' user experience and learning effectiveness.

By analyzing the use behavior and characteristics of learners, the entertainment robot can provide personalized folk art teaching services to meet the personalized needs of learners. By analyzing learners' search behavior and interest preference, the entertainment robot can accurately provide learners with the search results of folk art education resources. It can screen out the most relevant and suitable content from the massive folk art education resources according to the information needs of learners, and help learners quickly find the required learning materials. By analyzing learners' historical learning behaviors and interest preferences, the entertainment robot can recommend personalized folk art education content to learners. It can recommend learning resources, teaching activities and learning paths suitable for learners' needs and abilities according to their interests, learning levels and learning goals. This personalized recommendation can help learners better acquire the required folk art education knowledge, and improve the learning effect and learning motivation. Entertainment robots can also enhance the user experience and learning efficiency of learners through personalized user interface adjustments. It can adjust the display and interaction method of the system interface according to the learner's habits and feedback. For example, depending on the learner's preferences for different forms of media content, the entertainment robot can adjust the presentation of the interface to provide a more intuitive and easy-to-understand learning interface. This can improve learners' participation and learning effect, and further promote the personalized development of folk art education.

As shown in Fig. 3, the model provides users with a fast and accurate search and recommendation service for Folk fine art teaching resources by learning and tracking their personalized interests. By combining data analysis and machine learning techniques, the model can help users find content that meets their personalized needs in the vast online Folk fine art teaching library.

The user personality feature library can record personalized information of users in various ways. By analyzing user behavior and content, keywords that interest users can be extracted and saved in a personality feature library. These keywords can include keywords searched by users, clicked content tags, comments and favorite keywords, etc. By recording these keywords, users' interests and preferences can be reflected, providing a basis for subsequent personalized recommendations. The user personality feature library will continuously track changes in users' interests and update their personality features based on their new behaviors and feedback. For example, when users browse some new Folk fine art teaching resources and provide relevant feedback, the

Table 1

Experimental results of BP-KNN classifier.

Category	Recall factor	Precision ratio	F ₁ value
Woodcarving	80.910 %	84.669 %	82.106 %
Paper-cut for window decoration	76.136 %	81.426 %	78.899 %
Embroidery	72.820 %	76.748 %	74.404 %
Bird-and-flower painting	79.254 %	81.113 %	80.249 %
Clay sculpture	83.019 %	70.784 %	76.378 %
Dough modelling	86.249 %	78.074 %	82.179 %
Figure painting	89.165 %	82.702 %	86.148 %
Shehuo painting	63.378 %	79.960 %	70.771 %

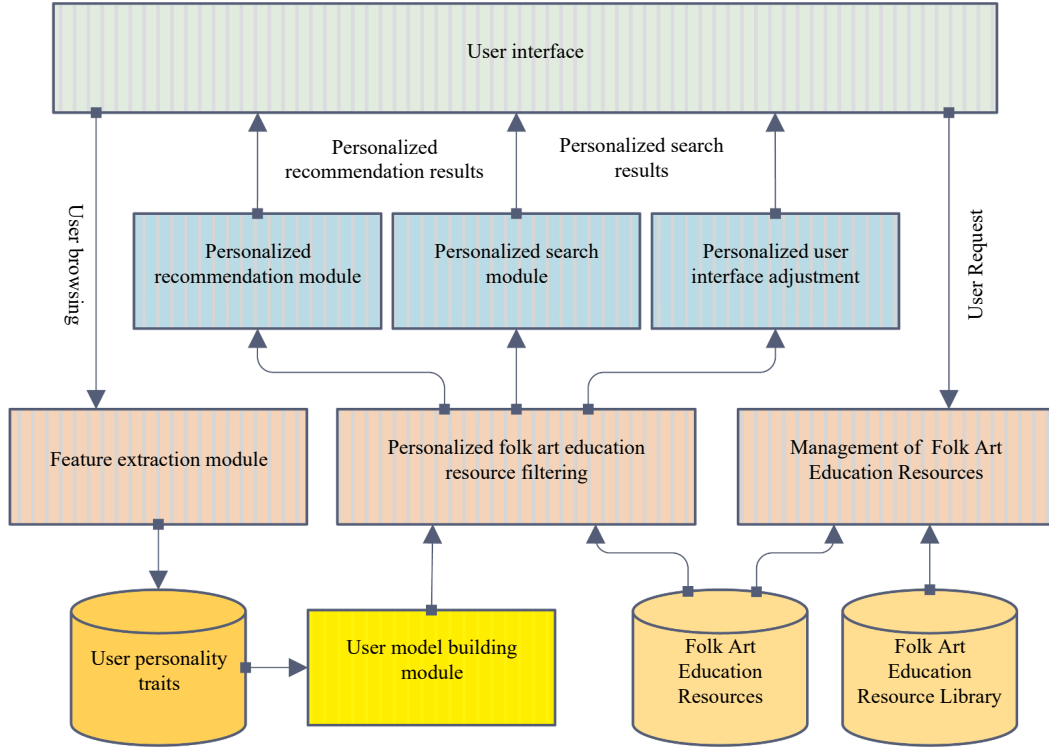


Fig. 3. Personalized Retrieval Model Structure.

personality feature library will update the user's interest features accordingly to maintain accurate tracking and recording of user interests. The user model construction module can establish user interest models based on the data in the user personality feature library, thereby achieving personalized recommendations and services. By analyzing the individual characteristics of users, the user model construction module can better understand their needs and preferences, and provide users with more accurate and personalized Folk fine art teaching recommendations.

Use the following formula (18) to calculate:

$$S_{ij} = \frac{\sum_{u \in U_{ij}} r_{ui} r_{uj}}{\sqrt{\sum_{u \in U_{ij}} r_{ui}^2 * \sum_{u \in U_{ij}} r_{uj}^2}} \quad (18)$$

In personalized recommendation, it is hoped to make recommendations based on the personalized needs of users. To achieve this goal, it is necessary to adjust the weight of users' rating behavior to accurately reflect their contribution to similarity. For active users, it is hoped to reduce their contribution to item rating similarity, while for non active users, it is hoped to increase their rating weight to enhance the accuracy of personalized recommendations. To achieve this goal, use formula (19) to calculate.

$$S_{ij-h} = \frac{\sum_{u \in U_{ij}} r_{ui} r_{uj} \frac{1}{\log(1+|N_u|)}}{\sqrt{\sum_{u \in U_{ij}} r_{ui}^2 * \sum_{u \in U_{ij}} r_{uj}^2}} \quad (19)$$

In a recommendation system, users' interests and preferences will change over time. In order to better adapt to this change, it is necessary to give users a penalty of time decay for items browsed over long time intervals to reduce their impact on similarity calculation. Use formula (20) to calculate:

$$S_{ij-f} = \frac{\sum_{u \in U_{ij}} r_{ui} r_{uj} \frac{1}{1 + \sqrt[2]{|t_{iu} - t_{ju}|}}}{\sqrt{\sum_{u \in U_{ij}} r_{ui}^2 * \sum_{u \in U_{ij}} r_{uj}^2}} \quad (20)$$

In order to simultaneously reduce the rating contribution of active users and punish the time decay of item ratings by users at different time periods, two optimization methods are combined to modify the calculation of item similarity and improve the accuracy of recommendation. Use formula (21) to calculate:

$$S_{ij-mix} = \frac{\sum_{u \in U_{ij}} r_{ui} r_{uj} \frac{2^{*(\log(1+|N_u|))^{-1}}}{1 + \sqrt[2]{|t_{iu} - t_{ju}|}}}{\sqrt{\sum_{u \in U_{ij}} r_{ui}^2 * \sum_{u \in U_{ij}} r_{uj}^2}} \quad (21)$$

By formula (21), when calculating similarity, both the attenuation penalty of user rating time and the activity of the item are considered. For items with longer time intervals or more active items, the corrected similarity will be lower to reduce their impact on recommendation results. For items with shorter time intervals or less active items, the corrected similarity will be higher to increase their impact on recommendation results. By combining these two optimization methods, we can more accurately calculate the similarity between items, thereby improving the accuracy of the recommendation system. In this way, whether it is reducing the rating contribution of active users or punishing the time decay of item ratings by users in different time periods, reasonable corrections can be made to adapt to changes in user interests and improve the effectiveness of recommendations.

4.2. Personalized transformation process of educational resources

Fig. 4 shows the process of user personalized interest feature transformation.

User personalized access features refer to the personalized features displayed by users when browsing, downloading, and evaluating Folk fine art teaching. These features can be extracted from user behavior

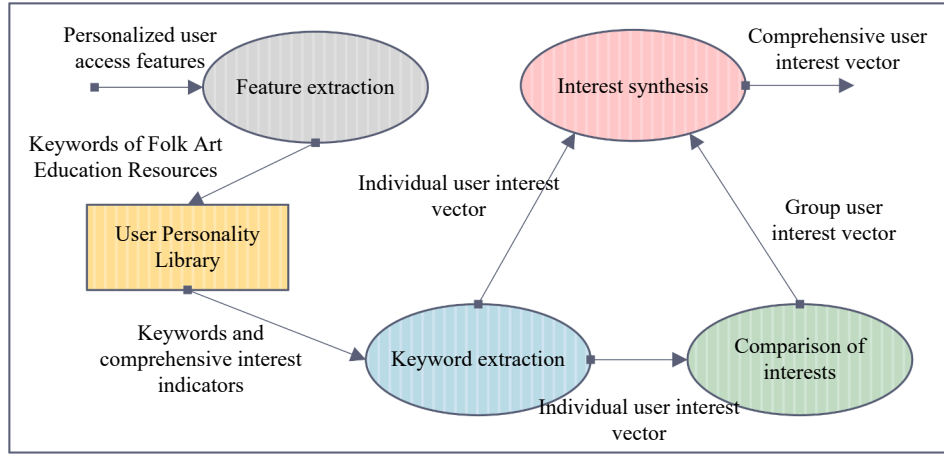


Fig. 4. User Personalized Interest Conversion Diagram.

data through feature extraction algorithms and saved in the user personalization library. In the process of feature extraction, the system can use various methods to identify the keywords of Folk fine art teaching that users are interested in. For example, natural language processing technology can be used to extract keywords and themes related to Folk fine art teaching by analyzing users' browsing records, download records, and evaluation content. These keywords can be terms related to artistic forms, artists, artistic styles, and specific content mentioned by users in evaluations. The extracted keywords can reflect users' interest and preferences in Folk fine art teaching, and can be saved in the user personalized library for subsequent personalized recommendations and matching. By analyzing the personalized access characteristics of users, the system can better understand their interests and needs, and provide users with more accurate and personalized recommendation services based on these characteristics.

The posterior probability of document d for type c_j can be calculated using formula (22):

$$p(c_j|d) = \frac{p(d|c_j)p(c_j)}{p(d)} \quad (22)$$

Here $P(d)$ represents the probability of document d , which can be estimated by the formula (23):

$$p(d) = \sum_{j=1}^n p(d|c_j)p(c_j) \quad (23)$$

And $P(c_j)$ can be estimated by formula (24):

$$p(c_j) = \frac{c_j \text{ Number of documents in}}{\text{Number of all documents in the document set}} \quad (24)$$

Assuming that the features of the document appear independently:

$$p(d|c_j) = \prod_{t \in d} p(t|c_j) \quad (25)$$

The estimated value is:

$$p(t|c_j) = \frac{n(c_j, t) + \lambda}{n(c_j) + \lambda|v|} \quad (26)$$

Update the conditional probability of each classification in the user interest model:

$$p(c_j|u) \leftarrow \frac{p(c_j|u) + \eta w_a p(c_j|d)}{1 + \eta w_a} \quad (27)$$

4.3. Comparison of the effectiveness of educational resource recommendation

In order to evaluate the performance of different algorithms, the classic association rule algorithm Apriori algorithm and the transitive closure clustering algorithm were selected as reference objects, and the algorithm proposed in this article was compared. The results of the evaluation experiment are presented in Tables 2 and 3.

According to the results in Table 3, it can be seen that our algorithm outperforms other algorithms in terms of recommendation accuracy.

These results indicate that our algorithm has higher accuracy and coverage in recommendation tasks, and can more accurately recommend learning resources that meet user interests.

4.4. Analysis of model training efficiency results

According to the results in Fig. 5, it can be seen that the comprehensive evaluation indicators for new user recommendations have changed in multiple rounds of repeated experimental testing.

According to the comprehensive evaluation index value F in Fig. 5, it can be concluded that the personalized resource recommendation model has relatively stable and ideal recommendation effects for new users in the system. This means that even for users who have never participated in resource scoring, the model can still provide relatively accurate and satisfactory resource recommendation services. This stable result indicates that the personalized resource recommendation model can effectively utilize information such as data, user characteristics, and resource attributes of other users when processing new users, thereby predicting the interests and preferences of new users. In this way, the model can provide resource recommendations that match the interests of new users, improving user satisfaction and user experience.

According to the results in Fig. 6, it can be seen that under the same experimental conditions, the personalized resource recommendation model and two other control experimental systems were trained and tested.

Table 2

Comparison of coverage when implementing web page recommendations.

Serial number	Apriori	Cluster	WRTC
1	37.385 %	35.575 %	38.791 %
2	42.288 %	42.167 %	43.399 %
3	45.357 %	47.687 %	48.276 %
4	46.457 %	48.311 %	50.900 %
5	47.812 %	49.445 %	51.646 %
6	50.901 %	51.748 %	53.904 %
7	56.811 %	57.503 %	59.616 %

Table 3
Comparison of Precision in Resource Recommendation Implementation.

Serial number	Apriori	Cluste	WRTC
1	73.754 %	68.882 %	77.761 %
2	68.837 %	65.167 %	77.397 %
3	57.480 %	56.705 %	70.138 %
4	55.292 %	51.671 %	68.690 %
5	50.391 %	48.618 %	63.504 %
6	46.861 %	43.747 %	59.442 %
7	46.562 %	43.090 %	59.035 %

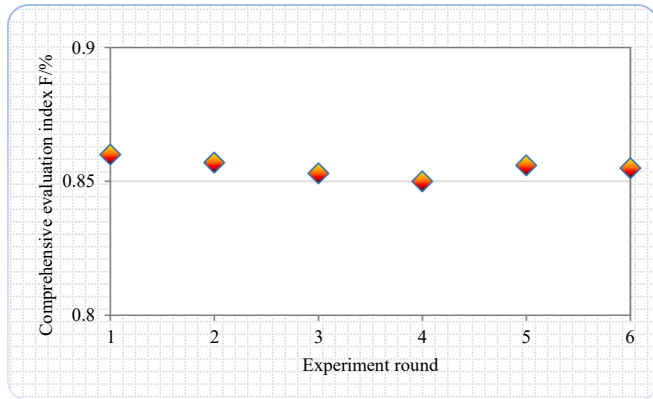


Fig. 5. Comprehensive evaluation indicators for new user recommendations.

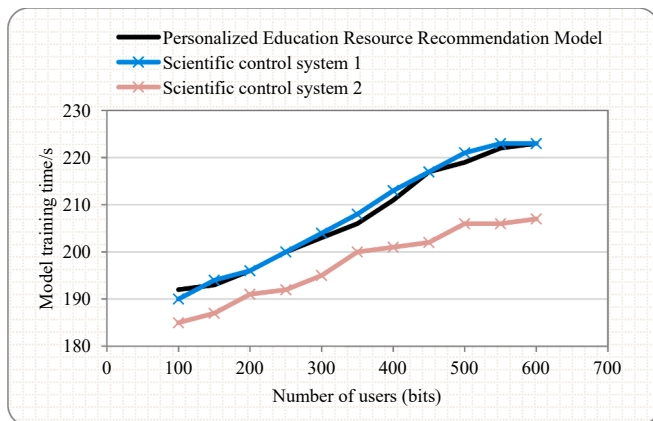


Fig. 6. Comparison of training time for three recommendation models.

The results in Fig. 6 indicate that the personalized resource recommendation model can efficiently utilize information such as user data and resource attributes during the training process, and quickly converge to the optimal solution. This model can provide accurate personalized recommendation services in real-time scenarios without excessively increasing the computational burden of the system.

5. Conclusion

Folk fine art arises from a rich and colorful life, reflecting the aesthetic taste of the working people and their desire for a better life. Art education can be divided into two meanings: broad and narrow. In a broad sense, art education refers to art education aimed at cultivating people's qualities and cultivation, that is, art education led by education; Narrowly defined art education refers to art education with the goal of cultivating specialized art talents, that is, art education led by art majors. This article mainly studies the teaching of art education in the university stage. Through data mining and information retrieval

technology, a personalized recommendation method for Folk fine art education is proposed, providing new ideas for Folk fine art education in China. Personalized recommendation methods can provide personalized art education resources and learning suggestions based on students' interests, abilities, and needs. Based on data mining technology, students' learning behavior and interest preferences can be analyzed to understand their preferences for different types of art works and artists. Through the interaction between entertainment robots and users, users' behavioral data can be collected, and natural language processing technology can be used to extract keywords and themes related to folk art teaching. At the same time, entertainment robots can also learn about users' preferences and preferences through questionnaires, game interaction and other ways. By analyzing users' personalized access characteristics, entertainment robots can better understand users' interests and needs, and provide users with accurate and personalized recommendation services. This personalized recommendation service based on entertainment robots can not only enrich users' learning experience and improve the effect of art education, but also increase users' dependence on and satisfaction with entertainment robots. At the same time, the personalized recommendation function of the entertainment robot can also help users quickly find the content most in line with their interests and preferences from the massive folk art teaching resources. By continuously optimizing and updating the personalized recommendation algorithm, the entertainment robot can provide users with more accurate and personalized art education recommendation services and promote the personalized development of art education. Through information retrieval technology, content related to students' interests can be screened from a vast amount of art education resources, providing personalized learning materials and art exhibition recommendations for them. This personalized recommendation method can help students better discover their interests and potential, improve learning motivation and effectiveness. It can also enrich students' art learning experience, allowing them to better appreciate the charm and uniqueness of Folk fine art. The application of this system can promote the development of Chinese Folk fine art education, provide students with more personalized learning resources, and meet their learning needs and interests. It can also provide data support for teachers and managers of Folk fine art education, better understand students' learning status and needs, and optimize teaching content and methods.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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